

With these meters, actions such as over-shooting your maximum demand (MD) control or introducing harmonics into the grid will trigger a response from the control centre directly to your meter (which is now more of a controller).

In smart meter chips, metrology is an analogue function where we accurately detect energy across a wide current range. "Integrating multiple analogue functions into one chip is a tremendously difficult task because it involves merging functions that were originally optimised in different process geometries. Ultimately, the biggest challenge is in making the performance of analogue functions similar or better," says David Andeen, segment manager—smart grid, Maxim Integrated.

One indigenous example of this technology is in the MREV e2o electric vehicle manufactured by Mahindra Reva Electric Vehicles Pvt Ltd. Stated as Car2Home technology, it allows to power your home by using a car's power during times of need. An on-board computer smartly regulates the power supplied to your home, and a full battery can ensure that the house is powered up for several hours depending on the load. This technology is complemented by the Sun2Car technology, in which a 10 sq m solar panel is set up to provide sufficient energy to power your car.

However, Diethelm says, "The electricity network is not designed for two-way power flows. This means that the network can experience issues when people generate power on their own and supply it to the grid. A common issue is higher than normal voltage levels when connecting solar system, which then damages appliances of consumers who are connected to the same grid."

Evolving communication standards. One of the challenges in this area is the lack of a standard protocol for communication. "Various state utilities are experimenting in a different manner, for instance Maharashtra has looked at Zigbee for AMR-related data collection," explains Thakurdesai.

"The current focus is on powerline communication (PLC), where we have

Factors that led to initiation of the July 2012 grid disturbance in India

1. Weak inter-regional corridors due to multiple outages
2. High loading on 400kV Bina-Gwalior-Agra link
3. Loss of 400kV Bina-Gwalior link

Ensuring grid security. To ensure grid security, islanding scheme for Delhi has been finalised and is under finalisation for UP, Punjab and Haryana. DISCOMs were asked to ensure compliance within $\pm 150\text{MW}$ or 12 per cent of their schedules irrespective of frequency. Feeder transmission lines have been identified for disconnection in case of violation of overpower withdrawal limits.

— *Report of Enquiry Committee on Grid Disturbance in Northern Region on July 30, 2012*

defined G3 and we are also implementing additional solutions. We have also partnered with RadioPulse for ZigBee solutions," adds Andeen.

Syam Madanapalli, CEO and co-founder, iRam Technologies, explains why a smart grid network is best based on IPv6 (the successor to IPv4; the next generation protocols for the Internet) and IEEE 802.15.4 (the wireless standard from IEEE for the Wireless Personal Area Networks (WPANs)). According to him, "Various devices at a given location in the proposed architecture form a mesh using IPv6 and IEEE 802.15.4 and communicate to rest of the smart grid using a border router. These meshes would be replicated throughout the electrical grid and finally connected to utility using appropriate wide area network (WAN) technologies that are commercially available from the telecom operators or ISPs."

Advanced metering infrastructure (AMI) and distribution automation. AMI is being more deeply integrated into the grid, and its projects are increasingly implemented by utilities, particularly for use in distribution automation (DA) applications.

AMI networks require robust communications between individual meters and data concentrators. They use either RF mesh or RF star topology, or implement PLC.

For example, to implement AMI, an IPv6-based wireless communication module is attached/integrated into the home energy meter. Few hundreds of homes in a given locality can form a mesh and connect to the utility using one-gateway routers. An additional border router can be provided for load

balancing and redundancy purpose.

Automated test equipment. The current slew of smart-grid test equipment is automated systems that can test all kinds of relays from manufacturers including ABB and Siemens, as well as different kinds of protection such as overcurrent, distance, differential and generator protection. When enquired about an instrument named smart Megger relay tester (SMRT), a senior engineer explained that conventional test equipment was fully manual and, as with any manual labour, there is a certain amount of human error that is introduced. Modern automated equipment takes away human error and also reduces the amount of work that he/she has to put in.

The voltage levels faced whilst testing circuit breakers are very dangerous. "Modern equipment allows you to do dual-ground testing, which enhances safety levels considerably. Conventionally, with both sides grounded, you are unable to even make a measurement, but our new solution is so unique in that this can be carried out," explains Ajay Goyal, managing director, Megger (India).

A switchgear will have electrical disconnect switches, fuses or circuit breakers used to control, protect and isolate electrical equipment. V. Narayanan, senior application engineer, Megger (India), says, "To check one switchgear, they needed the help of two to three people in order to test it properly. A new tool targeting this problem is the protection condition analyser. This tool allows to test simultaneously all of the key elements of substation protection systems—including protection relay,